

STUDENT ACADEMIC PERFORMANCE REPORT AND COURSE SYLLABUS

Student Information

Particulars	Details
Student Name	Gautham Manoj
Roll Number	AM.ID.R4IDS23007-FT
Academic Program & Branch	Ph.D. 2023
Current CGPA	8.76

Semester Performance Details

Semester 4

Semester	Course Code	Course Name	Academic Term Period	Course Type	Grade Obtained
4	22BIO819	Structural Biology	2024–2025 Even Semester	Regular	A

Performance Summary

Metric	Score
SGPA	9.0
CGPA	8.76

Course Details

Course Code: 22BIO819

Course Title: Structural Biology

Credits: 3-0-0-3

Grade Obtained: A

COURSE OVERVIEW

Structural Biology is an advanced interdisciplinary course offered by the Amrita School of Biotechnology, Amritapuri Campus, Amrita Vishwa Vidyapeetham. The course introduces students to the computational and structural aspects of biomolecules with emphasis on understanding sequence–structure–function relationships, three-dimensional protein structures, molecular docking, molecular dynamics simulations, and modern structural biology techniques. It combines theoretical knowledge with practical hands-on exercises and case studies, enabling students to apply computational approaches in biological research and drug discovery.

COURSE DESCRIPTION

The course includes the basic computational tools required to understand the structural aspects of biomolecules and enables students to apply these computational techniques in their research studies. It provides comprehensive exposure to structural databases, molecular visualization, docking studies, and molecular dynamics simulations for biological applications.

LEARNING OBJECTIVES

The course enables students to understand the importance of computational approaches in structural biology and modern biological research.

Students will gain knowledge of sequence–structure–function relationships and learn computational methods used to determine and analyze the three-dimensional structures of proteins and other biomolecules.

The course provides practical exposure to structural biology databases, visualization software, molecular docking techniques, and molecular dynamics simulations for understanding biomolecular interactions.

Students will also develop the ability to interpret structural data and apply computational methods for biomolecular design and research.

COURSE SYLLABUS

UNIT I – INTRODUCTION TO STRUCTURAL BIOLOGY

Introduction to Structural Biology, introduction to reading macromolecules, understanding the sequence–structure–function relationship, screening of target proteins and their structures, and hands-on practical exercises.

UNIT II – STRUCTURAL BIOLOGY DATABASES AND EXPERIMENTAL TECHNIQUES

Introduction to the Protein Data Bank (PDB) database, understanding and reading three-dimensional structure files, structural biology techniques including basics of macromolecular crystallography, Nuclear Magnetic Resonance (NMR), Cryo-Electron Microscopy (Cryo-EM), introduction to the RCSB database, reading PDB files, and related structural calculations.

UNIT III – MOLECULAR DOCKING AND LIGAND–PROTEIN INTERACTIONS

Introduction to ligand–protein interactions, concepts of molecular docking studies, introduction to small-molecule computational tools, molecular docking software, installation and use of AutoDock, preparation of ligand and protein files, importance of grid and docking files, preparation of grid parameters, preparation of docking parameters, and case studies.

UNIT IV – MOLECULAR DYNAMICS SIMULATION

Introduction to Molecular Dynamics (MD) Simulation, significance of simulation in biological research, building atomistic models, understanding atomic force fields, steps involved in molecular dynamics simulations, topology and parameter files, periodic boundary conditions, solvation and solvation methods, molecular simulation programs, and case studies.

PRACTICAL COMPONENTS

The course includes practical sessions on:

- Protein structure visualization
 - Reading and interpreting PDB files
 - Structural database analysis
 - Molecular docking using AutoDock
 - Ligand and protein preparation
 - Molecular dynamics simulation workflows
 - Structural analysis through computational tools
 - Case study discussions and research applications
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COURSE OUTCOMES

Upon successful completion of the course, students will be able to:

C01

Understand the fundamental concepts of Structural Biology and explain the sequence–structure–function relationship of biomolecules.

C02

Retrieve, visualize, and interpret macromolecular structures using the Protein Data Bank (PDB) and related computational tools.

C03

Apply computational techniques such as molecular docking to investigate ligand–protein interactions and analyze binding mechanisms.

C04

Perform molecular docking studies using AutoDock, prepare ligand and protein structures, generate docking parameters, and interpret docking results.

C05

Understand the principles of Molecular Dynamics Simulation and apply simulation techniques to investigate biomolecular stability, flexibility, and molecular interactions.

CO6

Utilize computational structural biology approaches in biomolecular research, drug discovery, protein engineering, and related biological applications.

COURSE APPLICATIONS

The knowledge acquired through this course can be applied in:

- Structural Bioinformatics
 - Drug Discovery and Drug Design
 - Computational Biology
 - Protein Engineering
 - Molecular Modeling
 - Molecular Docking
 - Molecular Dynamics Simulation
 - Pharmaceutical Research
 - Biotechnology
 - Systems Biology
 - Precision Medicine
 - Academic and Industrial Research
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COURSE RESULT SUMMARY

Course Code	Course Title	Semester	Credits	Grade	SGPA
22BIO819	Structural Biology	4	3-0-0-3	A	9.0

ACADEMIC RECORD

Student Name: Gautham Manoj

Roll Number: AM.ID.R4IDS23007-FT

Programme: Ph.D. 2023

Course Completed: Structural Biology (22BIO819)

Semester: 4

Grade Awarded: A

Semester GPA (SGPA): 9.0

Cumulative GPA (CGPA): 8.76

Academic Term: 2024–2025 Even Semester

Prepared as a consolidated academic record containing the student's performance details and complete syllabus for 22BIO819 – Structural Biology.